

Build Non-Contact Type DIGITAL TACHOMETER

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A tachometer measures the rotation speed of motors and other machineries. There are various types of tachometers. Described here is a digital non-contact type tachometer using a proximity sensor.

Circuit and working

The circuit diagram of the Arduino-based tachometer is shown in Fig. 1. It is built around Arduino Uno board (Board1), inductive pnp-type normally-open (NO) proximity sensor, 16x2 LCD display (LCD1), optocoupler PC817 (IC1) and a few other components.

A metal such as a bolt head is fixed on the shaft of the rotating object. Whenever the metal crosses the sensor during rotation, the sensor generates a pulse at its output. The program in Arduino counts the number of pulses received for one second and converts into rotations per minute (RPM). The rpm count is updated every second and is displayed on LCD1.

Proximity sensor. A proximity sensor detects objects nearby without any physical contact up to its nominal range. The maximum distance that the sensor can detect is its nominal range. A proximity sensor can have high reliability and long functional life because of the absence of mechanical parts and physical contact between the sensor and the target/object. A pnp-type NO proximity sensor used in this project is shown in Fig. 2.

The program (tachometer.ino) is written in Arduino programming language. Arduino IDE is used to compile and upload the program to Arduino board.

Select the correct board from Board→Tools menu in Arduino IDE, select COM port and upload the program (tachometer.ino) through the standard USB port of the computer.

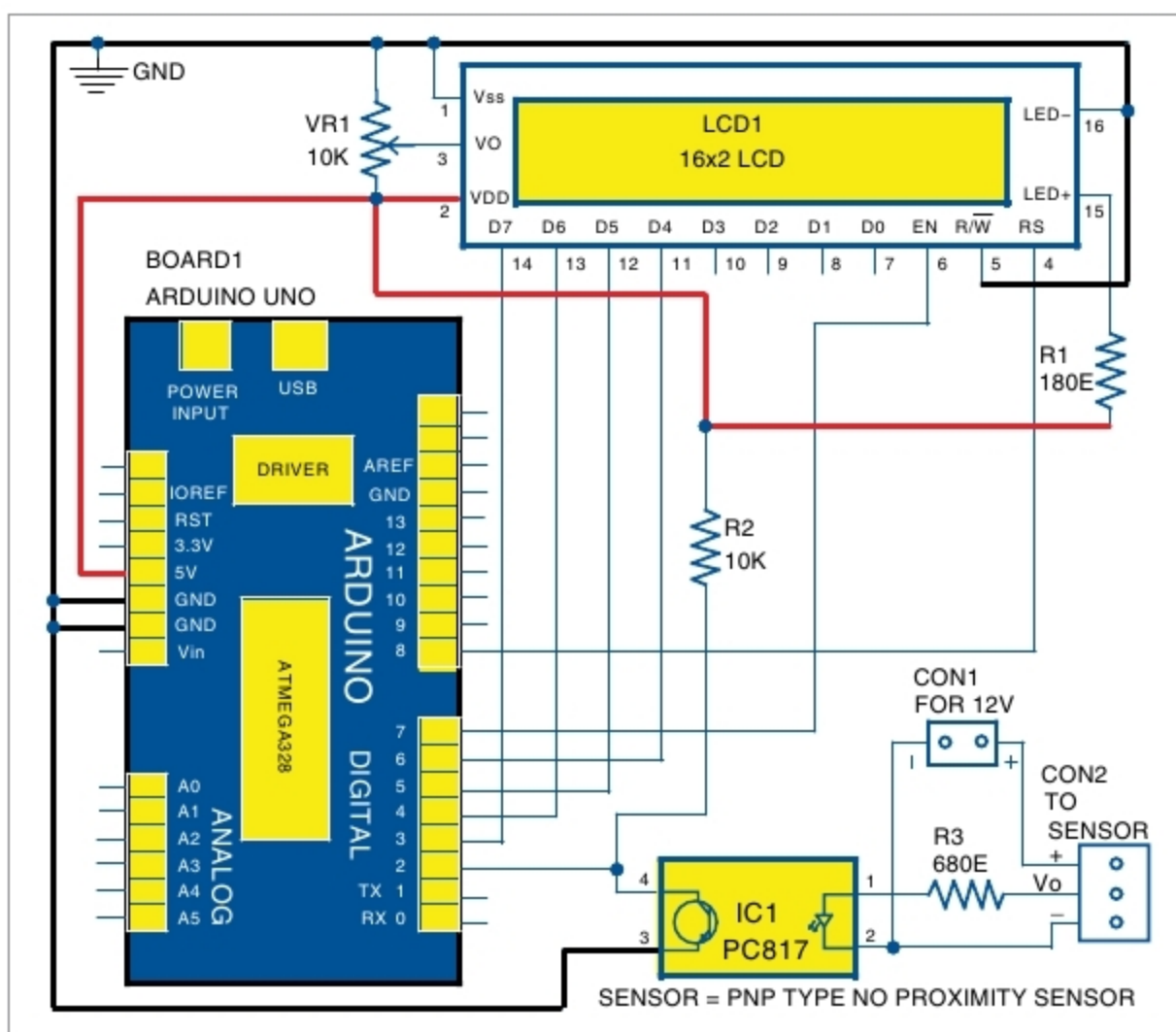


Fig. 1: Circuit diagram of Arduino-based tachometer



Fig. 2: A pnp-type proximity sensor

Construction and testing

An actual-size PCB layout of the Arduino-based tachometer is shown in Fig. 3 and its components layout in Fig. 4. Assemble the circuit on the PCB. Connect Arduino Uno to the PCB using external jumper wires. Power

supply for Board1 is provided by the computer's USB port.

Sensor setup. After assembling the circuit on the PCB, connect output of the proximity sensor across connector CON2 and connect an external 12V DC

PARTS LIST

Semiconductors:

- IC1 - PIC817 optocoupler
- BOARD1 - Arduino Uno board
- LCD1 - 16x2 LCD

Resistors (all 1/4-watt, ±5% carbon):

- R1 - 180-ohm
- R2 - 10-kilo-ohm
- R3 - 680-ohm
- VR1 - 10-kilo-ohm potmeter

Miscellaneous:

- CON1 - 2-pin connector
- CON2 - 3-pin connector
- SENSOR - pnp-type NO proximity sensor
- Jumper wires
- 12V DC power supply